

Terminal Operator

A Lock-Aware Right-Truncation Operator for Recursive Systems

Classification: Operator Theory / Recursive System

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Abstract

We define a non-linear, lock-aware right-truncation operator T acting on the space of natural numbers $N_{\geq 4}$. T projects an arbitrary state s onto a discrete lattice of admissible points $L = 4N_0$. By allowing a restricted local perturbation $\delta \in \{0,1\}$, we prove that the operator induces a strictly contractive sequence that terminates in finite steps. We further characterize the set of "Terminal States" where $T(s) = \perp$, denoting a configuration where no admissible projection exists within the allowed degrees of freedom.

1. Definitions and Domain

The Law of Admissibility ($R = 4$) establishes that stability in recursive systems is maintained through alignment at modular checkpoints. We define the lock set L as:

$$L = \{ 4k \mid k \in N_0 \}$$

For any state $s \in N_{\geq 4}$, we define the Floor Projection k_s and its Unit Successor k_s^+ :

$$k_s = \lfloor s/10 \rfloor, \quad k_s^+ = \lfloor s/10 \rfloor + 1$$

The choice of base-10 truncation functions as a resolution filter, systematically reducing the state space by one digit per application. Because the admissible lattice $4N_0$ has a period of 4 and the truncation operates with a period of 10, the interaction between these two periodicities creates a structured pattern of recoverable and terminal states. The greatest common divisor $\gcd(10,4) = 2$ ensures that the sampling of the lock set by the truncation operator produces a repeating 20-unit cycle, with alternating admissible and terminal blocks. This is not a signal processing operation but a structural constraint on state reduction.

2. The Operator T

The operator $T: N_{\geq 4} \rightarrow L \cup \{\perp\}$ is defined by the selection function:

$T(s) =$	k_s	if $k_s \in L$
	k_s^+	if $k_s \notin L$ and $k_s^+ \in L$
	$\perp$	otherwise

The degree of freedom is strictly bounded by the perturbation $\delta = T(s) - k_s$, where $\delta \in \{0, 1\}$. This represents the minimum unit of freedom required to maintain stability in a recursive transition.

3. Stability and Convergence

Theorem 1 (Strict Contraction). For all $s \in \text{dom}(T)$, $T(s) < s$.

Proof for $s \geq 40$: For $s \geq 40$, $k_s = \lfloor s/10 \rfloor \leq s/10$. The maximum possible value for $T(s)$ is $k_s + 1$. Since $s/10 + 1 < s$ for all $s > 10/9 \approx 1.11$, and $40 > 1.11$, the inequality $T(s) < s$ is strictly satisfied.

Proof for $4 \leq s < 40$: Manual verification confirms that for all $s \in [4, 39]$, $T(s) \in \{0, 4\}$, both of which are strictly less than s . $\square$

Theorem 2 (Finite Termination). Every sequence $s_{i+1} = T(s_i)$ terminates in $N \leq \log_{10}(s_0)$ steps at either a stable lock point in L or the null-state $\perp$.

Proof. By Theorem 1, each step strictly decreases the value. The number of steps is bounded by the number of digits, hence $N \leq \log_{10}(s_0)$. $\square$

4. Classification of Terminal States

A state t is Terminal if the neighborhood of its floor projection contains no elements of the lock set L :

$$T(t) = \perp \iff \{ \lfloor t/10 \rfloor, \lfloor t/10 \rfloor + 1 \} \cap 4N_0 = \emptyset$$

Admissibility Predictor Function. For any state s , let $a = \lfloor s/10 \rfloor$. The state is Recoverable if:

$$(a \bmod 4 = 0) \text{ or } ((a+1) \bmod 4 = 0)$$

Otherwise, s lies in a Terminal Block.

Because the lock points in $4N_0$ are spaced at intervals of 4, and the operator's reach (k_s, k_s^+) covers only 2 consecutive integers, "Admissibility Gaps" appear periodically in the number line. The interaction between base-10 truncation and 4-periodic locks produces a repeating 20-unit cycle, with 10 units admissible and 10 units terminal.

5. Applications in Quantum Architecture

In n -gate quantum arrays, we define the Vector Terminal Operator T^n acting component-wise.

Definition (Terminal Drift). Terminal Drift is the cascading propagation of inadmissible states through interconnected components. In high-dimensional systems, a single component reaching $\perp$ necessitates a global Halt. This "Hard Reset" prevents the corruption of the global state vector, ensuring that the machine operates as a Glass Box — either perfectly aligned or fundamentally inert.

The operator provides a deterministic reset mechanism for the Blossom Harmonic Algorithm (BHA), ensuring that any system reset lands on a known admissible lock point without introducing floating-point drift or recursive error accumulation.

Appendix: Admissibility Map (s < 200)

The following table classifies the first 200 natural numbers under the Terminal Operator.

Range s	Floor k_s	Lock Target T(s)	Status
4 – 49	0 – 4	0 or 4	Admissible
50 – 69	5 – 6	⊥	Terminal Block
70 – 89	7 – 8	8	Admissible
90 – 109	9 – 10	⊥	Terminal Block
110 – 129	11 – 12	12	Admissible
130 – 149	13 – 14	⊥	Terminal Block
150 – 169	15 – 16	16	Admissible
170 – 189	17 – 18	⊥	Terminal Block
190 – 199	19 – 19	20	Admissible

Notes:

- For range 4-49, floor values 0-4 map to lock points: 0 (trivial base) and 4.
- For range 70-89, floor 7-8 map to lock point 8.
- For range 110-129, floor 11-12 map to lock point 12.
- For range 150-169, floor 15-16 map to lock point 16.
- For range 190-199, floor 19 maps to lock point 20.
- The pattern repeats every 20 units: 10 units admissible, 10 units terminal block, with the admissible window centered on multiples of 4. This 20-cycle structure is a direct consequence of $\gcd(10,4) = 2$ and the Law of Admissibility.

References

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